



Amir Hossein Madadi

B.Sc. Electrical Engineering,
University of Tehran — Power Systems

Final-year Electrical Engineering student specializing in power systems, electrical machines, transformers, and engineering simulation. Proficient in computational analysis, algorithm implementation in MATLAB, and digital system design with Verilog HDL.

Contact

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Technical Skills

Power Systems & Machines

- MATLAB
- Simulink
- Ansys Maxwell
- Simcenter SPEED
- COMSOL
- DlgSILENT PowerFactory
- PowerWorld Simulator

Circuits, Electronics & Design

- PSpice
- LTspice
- Multisim
- OrCAD Capture
- Altium Designer
- AutoCAD
- PCB Design

Digital, Control & Programming

- Verilog HDL
- ModelSim
- Quartus Prime
- Python
- C / C++
- Siemens PLC

Languages

Persian: Native
English: Advanced
Arabic: Intermediate

Interests

Electrical machines & transformers
Power grid analysis & operation
Motor control & dynamic simulation
Digital design & processor architecture

Seeking internship or collaboration opportunities in electrical engineering projects.

Education

B.Sc. in Electrical Engineering

2021 — Present

University of Tehran, Faculty of ECE — Power Systems

Selected Projects

Power System Analysis & Load Flow

- Load flow and short-circuit studies on the IEEE 39-Bus test network using DlgSILENT PowerFactory, including maximum loadability, generator outage scenarios, and voltage profiling.
- N-1 Contingency analysis and three-phase fault current studies on the test network.
- 400kV/230kV transmission line parameter analysis on real tower configurations and network operation under various loading conditions.
- Implementation of Newton-Raphson, Fast Decoupled, and Gauss-Seidel algorithms in MATLAB for convergence comparison on large networks.

DlgSILENT PowerFactory MATLAB IEEE 39-Bus Load Flow Short Circuit

High-Voltage Insulation & Electric Field Simulation

- 2D electric potential and field distribution analysis between electrodes using Finite Difference Method (FDM) and Monte Carlo simulation in MATLAB.
- Effect of multiple dielectric layers with varying permittivity and air voids analyzed using COMSOL.
- Traveling wave simulation on 300Ω overhead lines (1600km/400km) with damping behavior studied in LTspice.

COMSOL MATLAB LTspice FDM Monte Carlo

Electrical Machines & Transformer Design

- Design and construction of a high-frequency transformer using EE55 ferrite core, including core selection, litz wire for skin-effect mitigation, and practical validation.
- Three-phase 7.5kW/380V induction motor simulation in Ansys Maxwell-RMxprt with stator/rotor characterization and torque-speed analysis.
- DC motor design and optimization in Simcenter SPEED including geometric and electromagnetic parameter calculation.

Ansys Maxwell RMxprt Simcenter SPEED COMSOL

Control Systems & Dynamic Modeling

- DC shunt motor speed control implementation and performance analysis in Simulink/Simscape.
- Physical modeling of a single-loop DC motor including magnetic field, electromagnetic force, and induced EMF in MATLAB.
- CSTR (irreversible exothermic A→B reaction) modeling and control with 3 coupled differential equations in Simulink.

Simulink Simscape MATLAB CSTR

FPGA Digital System Design

- Direct Digital Synthesis (DDS) function generator on FPGA with DAC-PWM, generating sine, square, triangular, sawtooth, half-wave, and full-wave rectified waveforms.
- Radix-4 multiplier and Exponential Engine SoC accelerator implementation in SystemVerilog.
- Fundamental FPGA circuits including ring oscillator, frequency divider, and Baud rate generator.

Verilog Quartus Prime FPGA SystemVerilog

RISC-V Processor Design

- Design and simulation of three RV32I-class processor architectures (single-cycle, multi-cycle, and 5-stage pipeline) in Verilog HDL.
- Control unit, 32-register file, 32-bit ALU, and instruction/data memory implementation.
- Data and control hazard management: Forwarding, Stall, Flush for 5-stage pipeline.
- Validation with RISC-V assembly programs including 32-bit signed array sorting.

Verilog HDL ModelSim Quartus Prime RISC-V